

A yellow LEGO Technic bulldozer model is shown in the background, built on a base plate. The bulldozer has a large yellow blade, a black rubber track, and various mechanical components like gears and axles. The text is overlaid on this image.

CS 180 Discussion 9

Neural Radiance Field (NeRF) Fundamentals

Agenda

1. What is a NeRF “neural radiance fields”?
2. How do we train a NeRF?
 - a. Positional encoding
 - b. Volumetric rendering

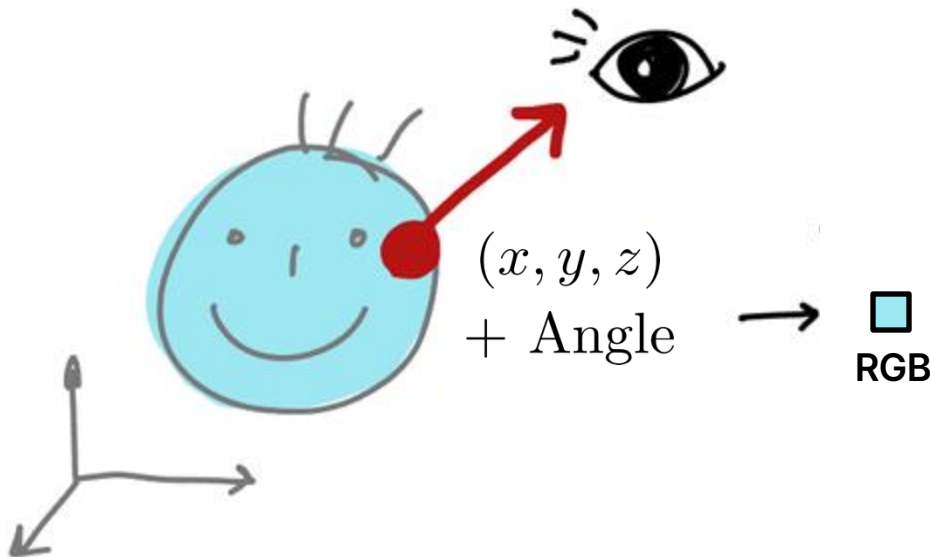
What is a NeRF?

“Neural Radiance Fields”

What is a NeRF?

"Neural Radiance Fields"

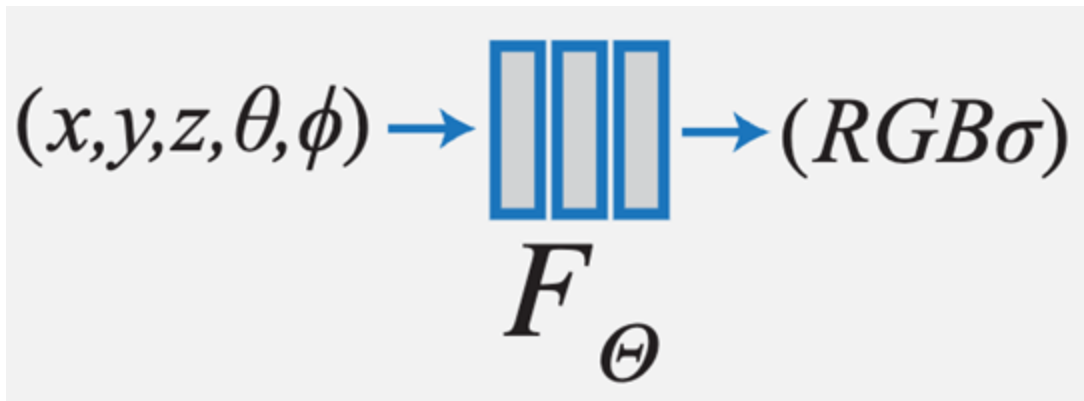
- Radiance Fields: point (+ viewing angle) $\rightarrow$ color



What is a NeRF?

"Neural Radiance Fields"

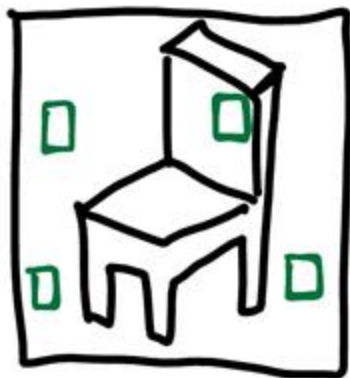
- Radiance Fields: point (+ viewing angle) $\rightarrow$ color
- Neural: use neural networks to learn this mapping
 - Supervised learning: (input) $\rightarrow$ (output: predicted value) $\leftrightarrow$ (actual value)



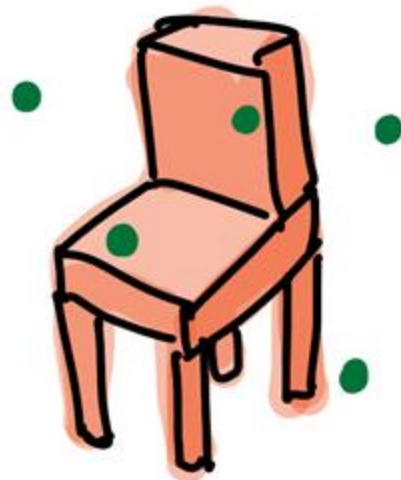
How do we train a NeRF?

- We can learn neural images (2D) or neural radiance fields (3D)

Supervise 2D pixels



Supervise 3D points



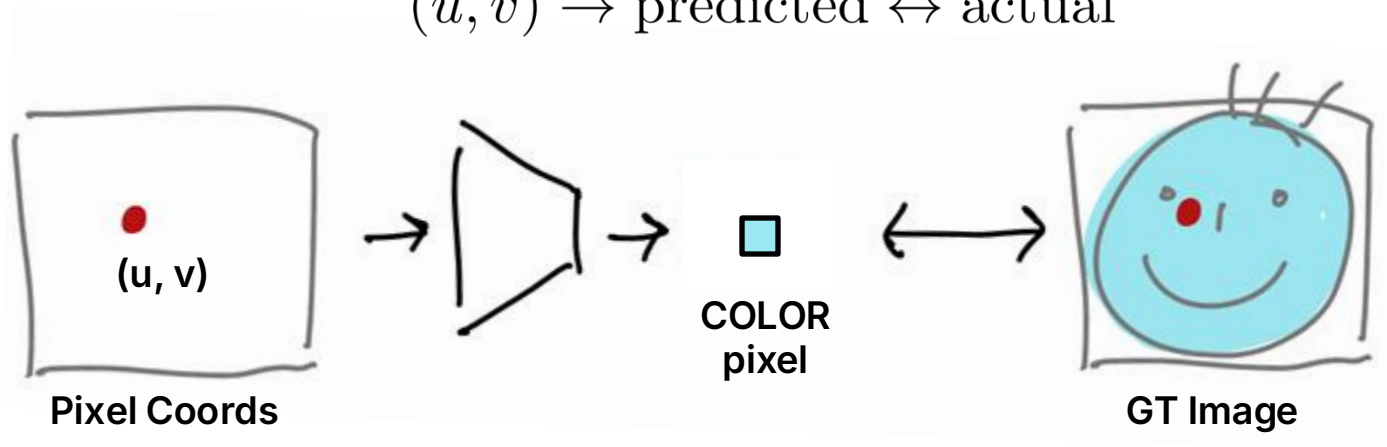
Training a neural image

Input: pixel coordinates (u, v)

Output: color (RGB)

Ray supervision :-)

$(u, v) \rightarrow \text{predicted} \leftrightarrow \text{actual}$



Training a neural image

Input: pixel coordinates (u, v)

Output: color (RGB)

Does it work well?

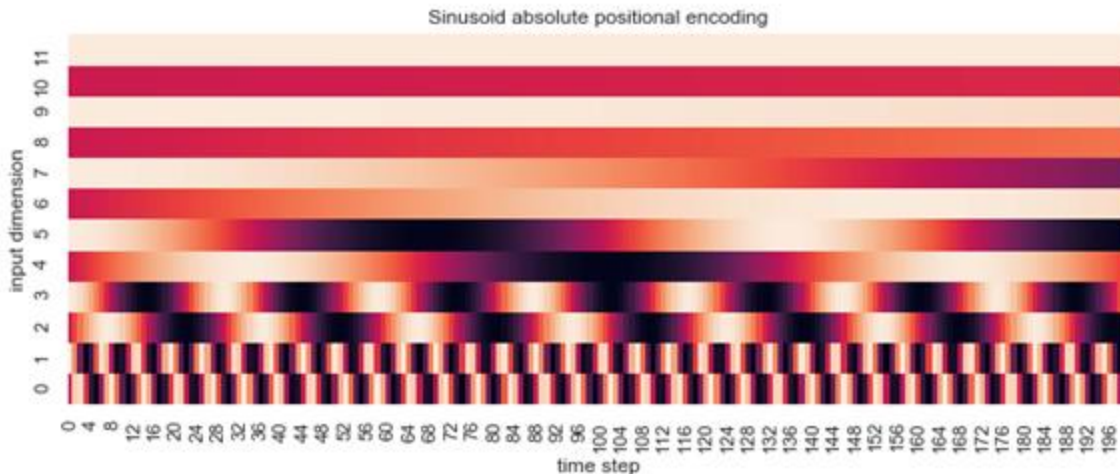
We need
something more ->



Positional Encoding

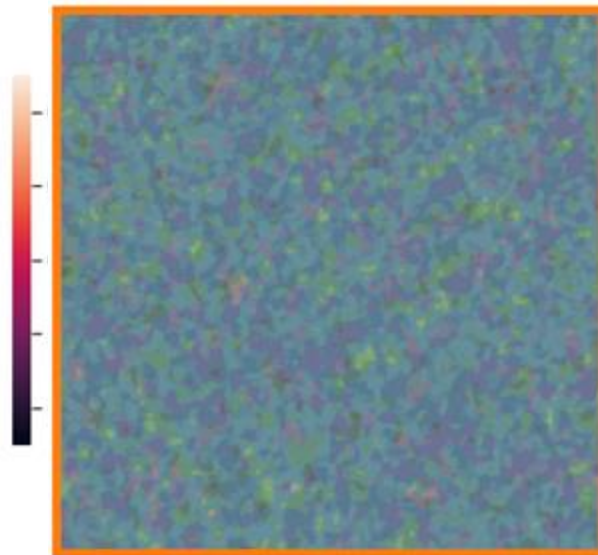
$$(u, v) \rightarrow [PE(u), PE(v)] \rightarrow \text{NN}$$

Helps model sharp changes (why?)



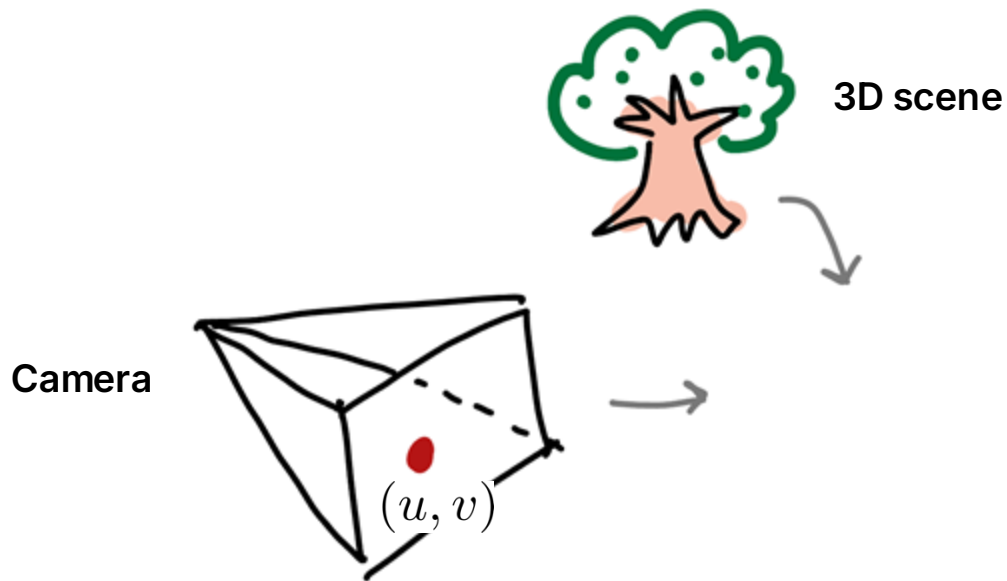
from 16 to 20

With PE

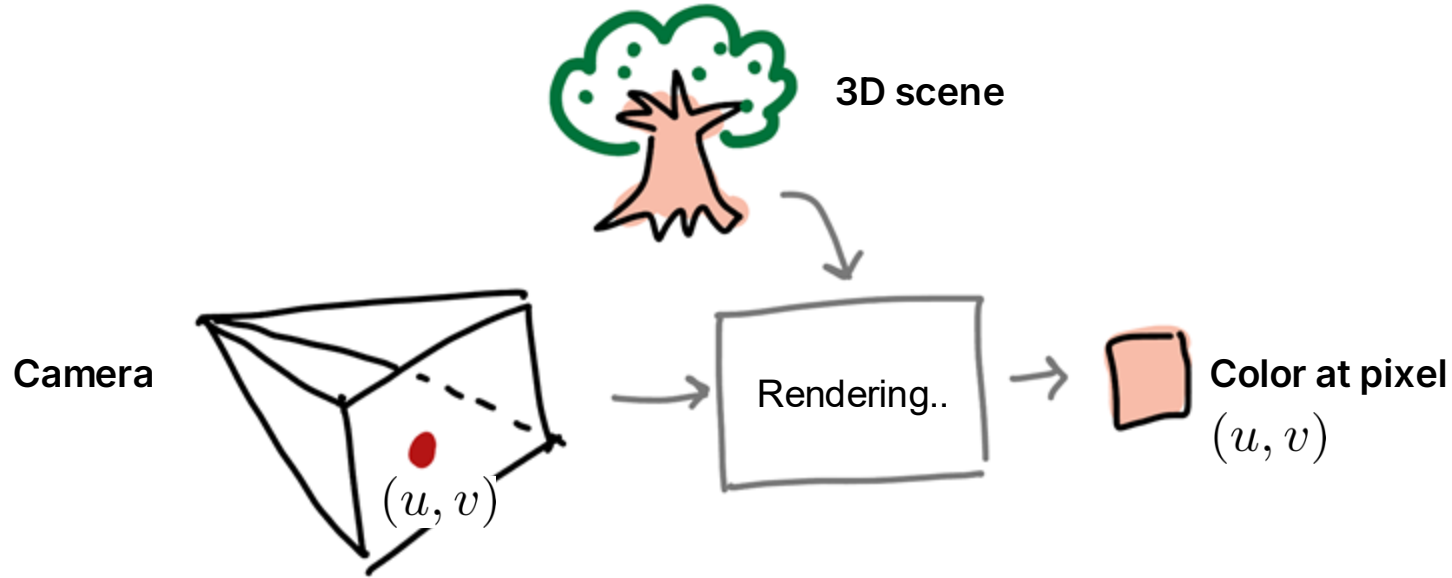


Training NeRF

Volumetric Rendering (Philosophy-level)



Volumetric Rendering (Philosophy-level)



Training a NeRF

Input: (camera) + (u,v)

Output: color (RGB)

What do we have?

Training a NeRF

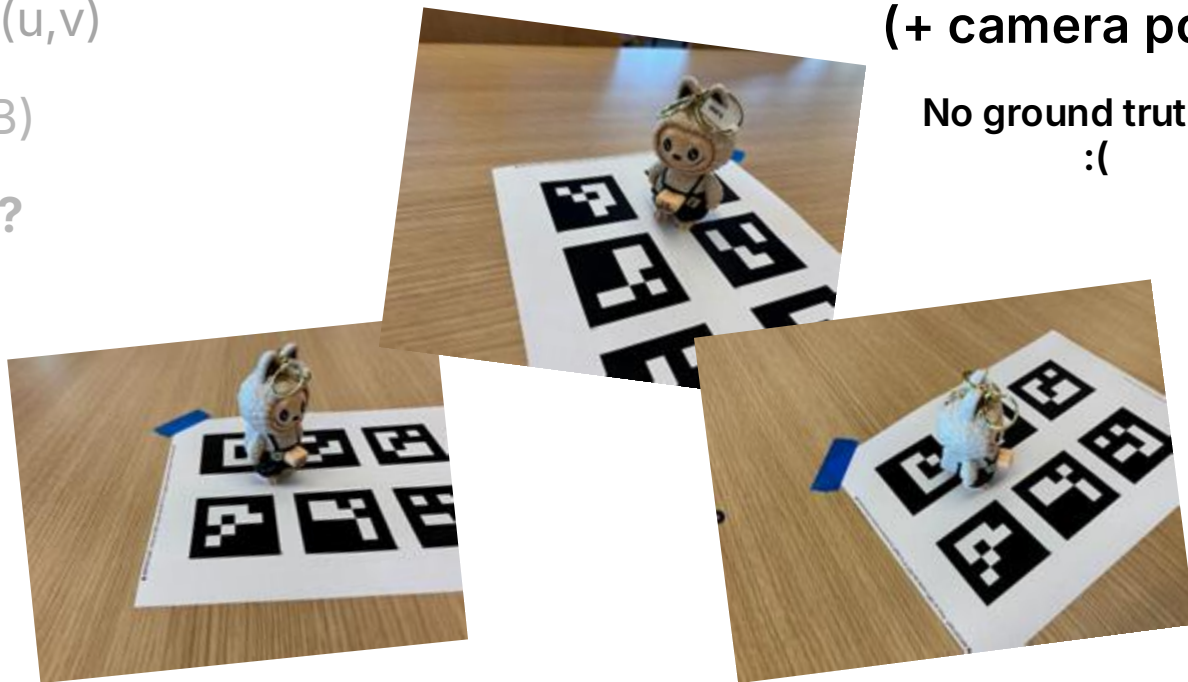
Input: (camera) + (u,v)

Output: color (RGB)

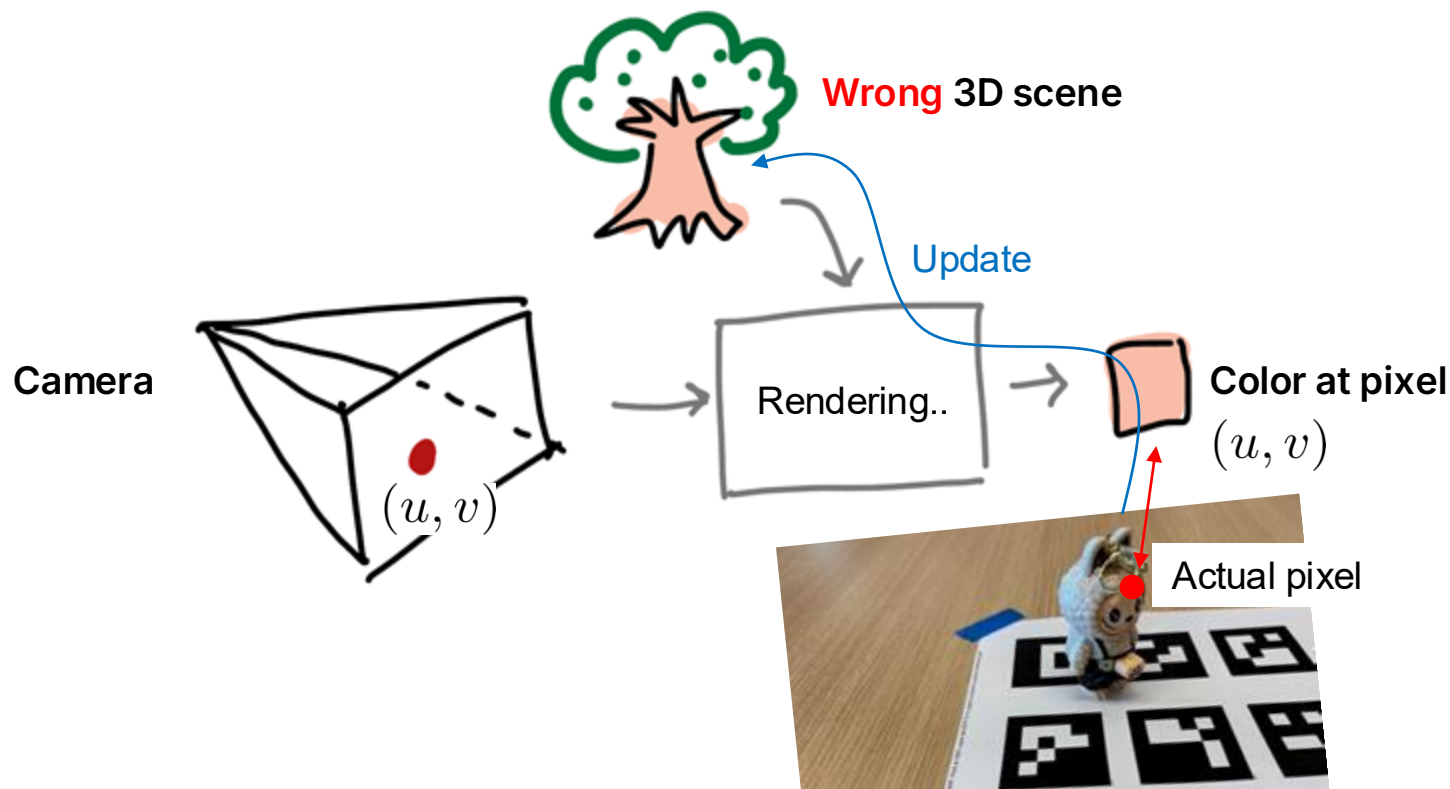
What do we have?

Images
(+ camera poses)

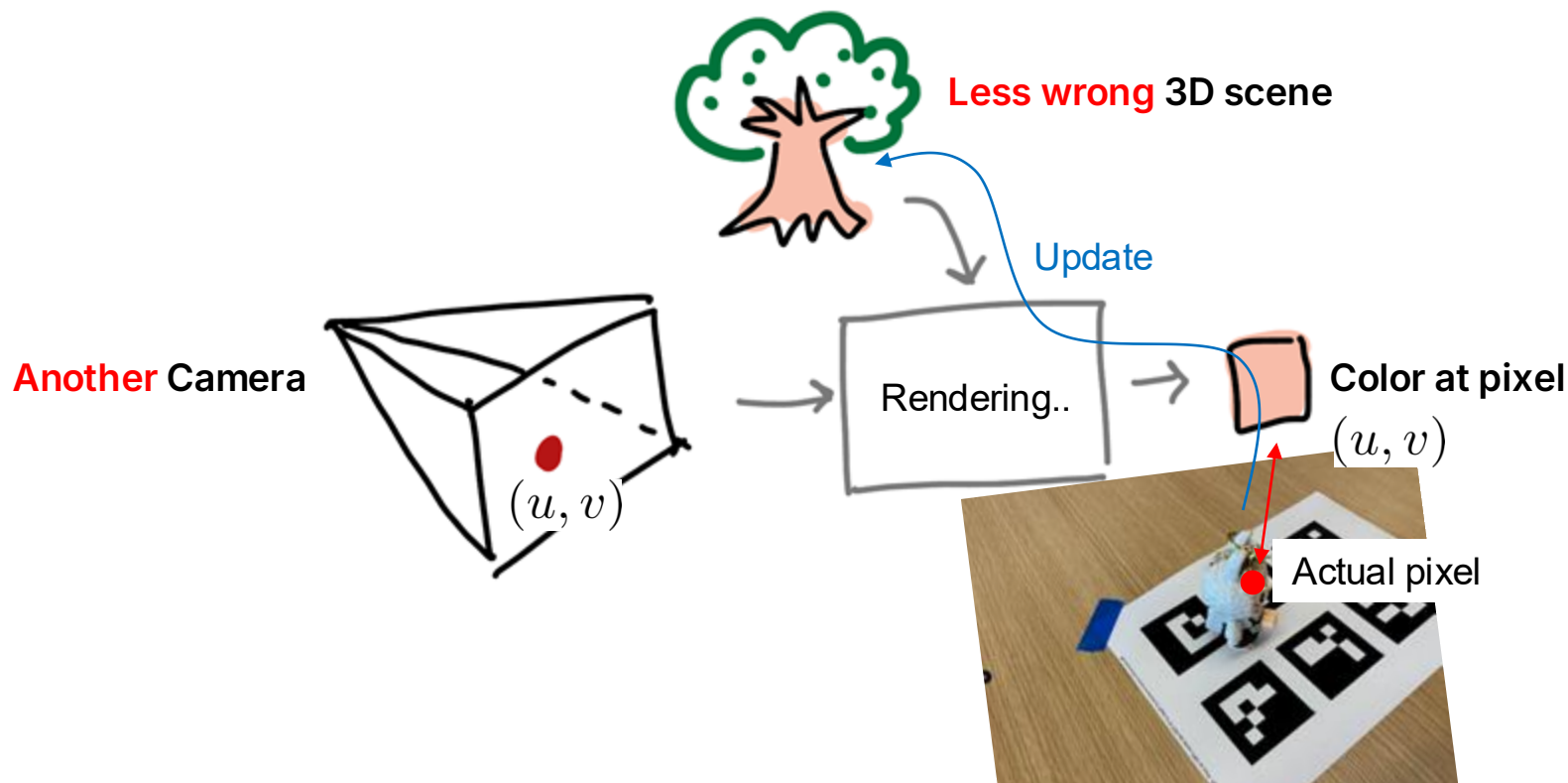
No ground truth 3D
:(



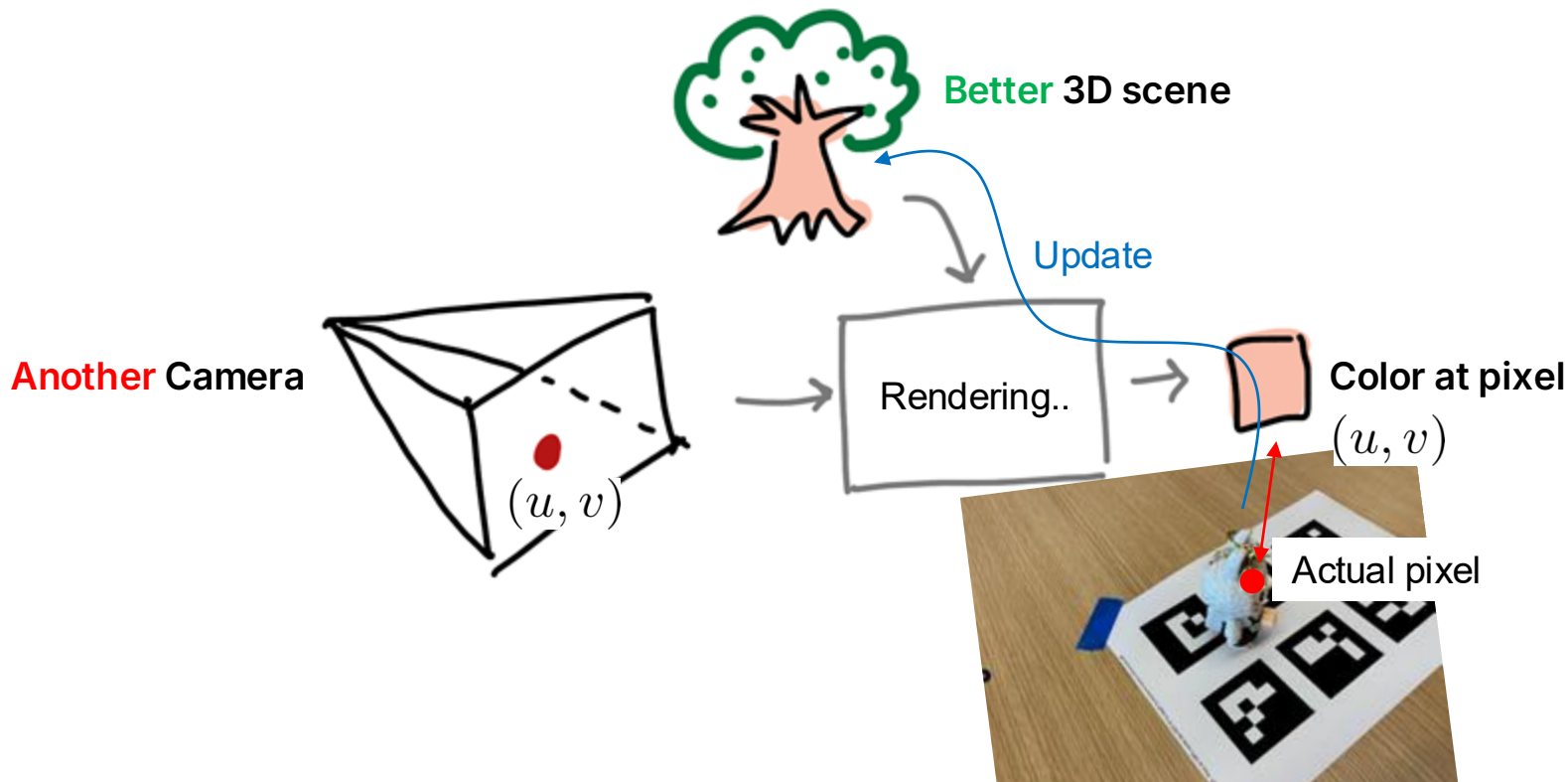
Training a NeRF (Philosophy-level)



Training a NeRF (Philosophy-level)



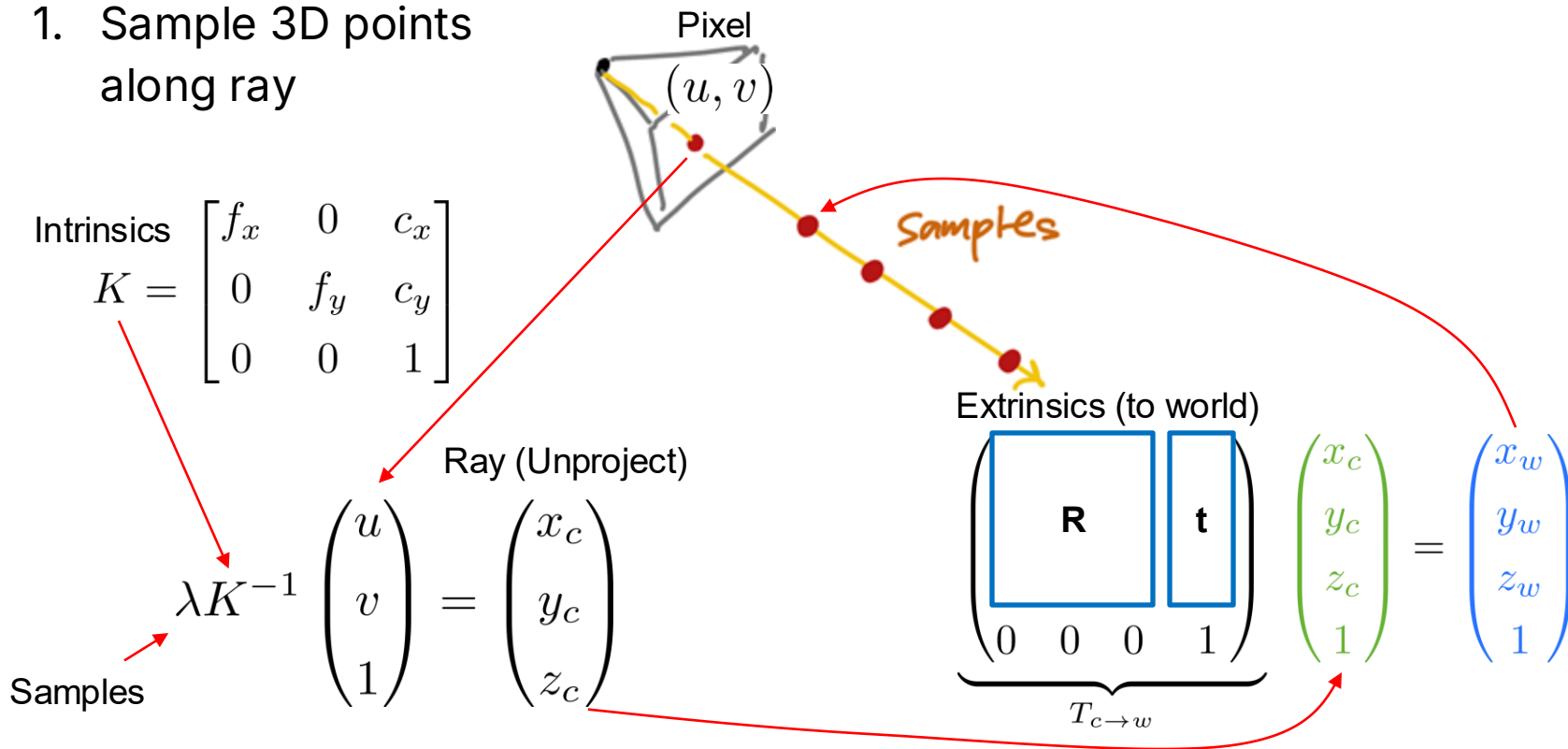
Training a NeRF (Philosophy-level)



How NeRF renders?

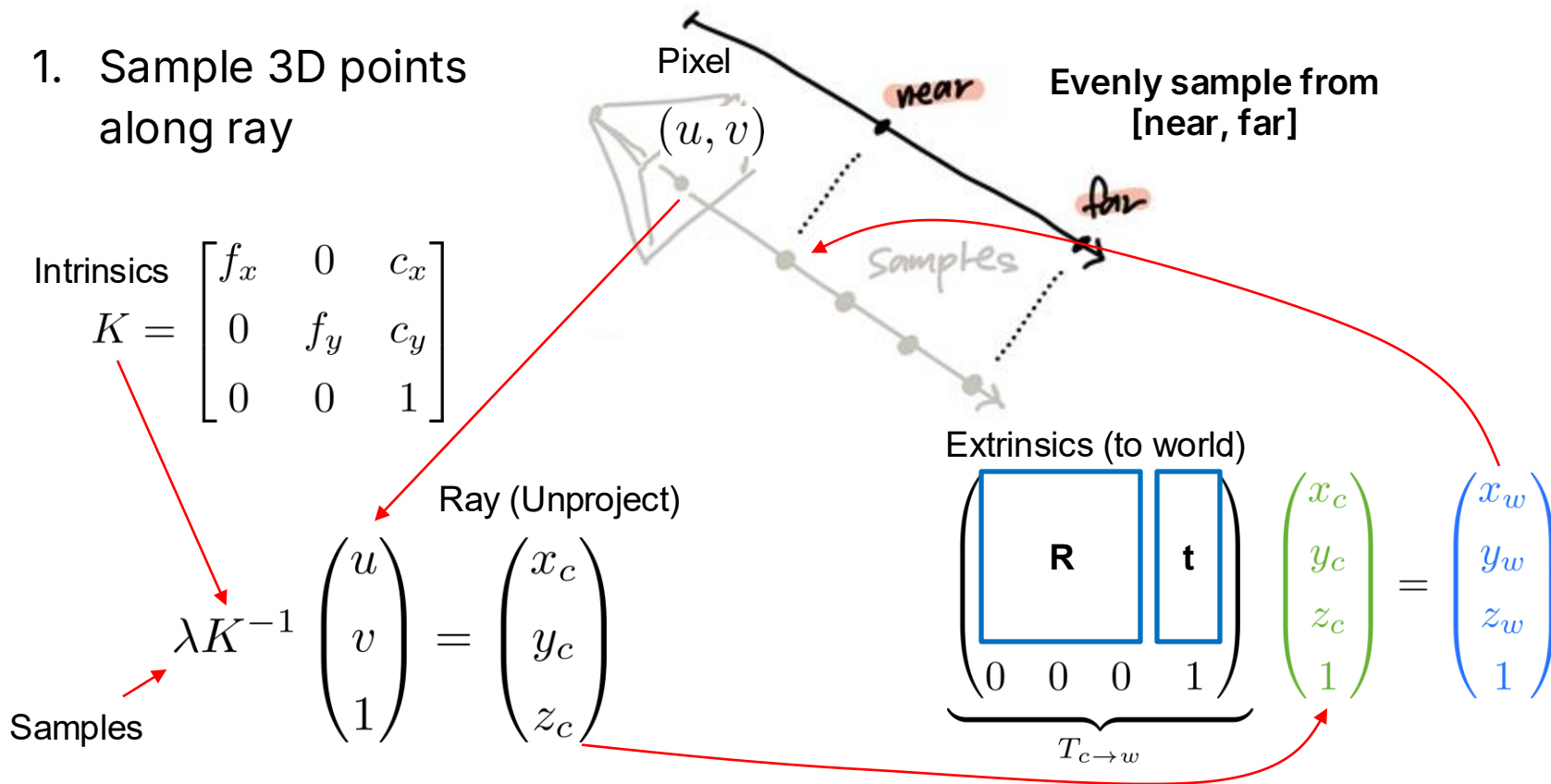
How NeRF renders? (3 steps)

1. Sample 3D points along ray



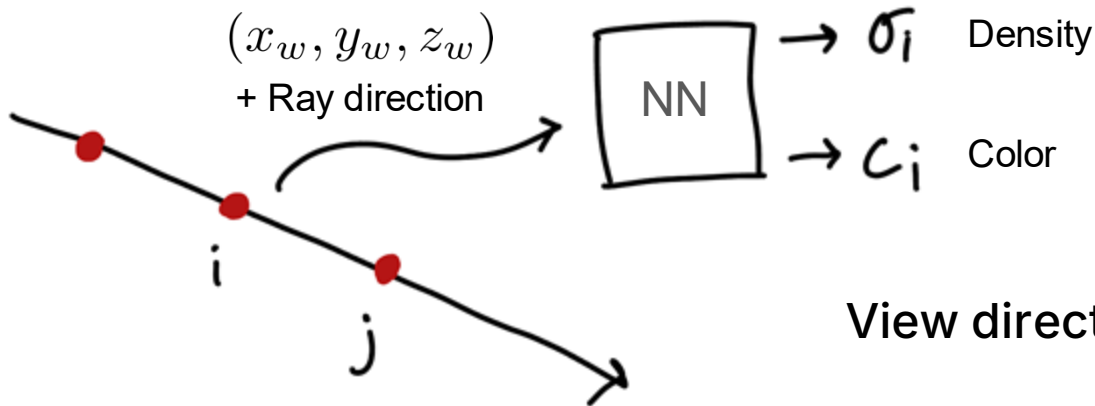
How NeRF renders?

1. Sample 3D points along ray



How NeRF renders?

1. Sample 3D points along ray
2. Predict (color, density) at each 3D point
3. Render color at 2D pixel (using volumetric rendering)



View direction $\leftrightarrow$ Ray direction

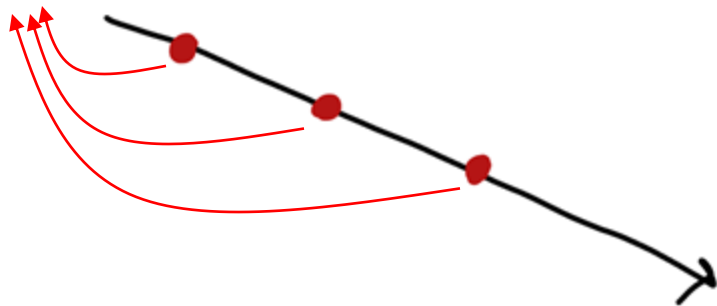
How NeRF renders?

1. Sample 3D points along ray
2. Predict (color, density) at each 3D point
3. Render color at 2D pixel (combine using volumetric rendering)

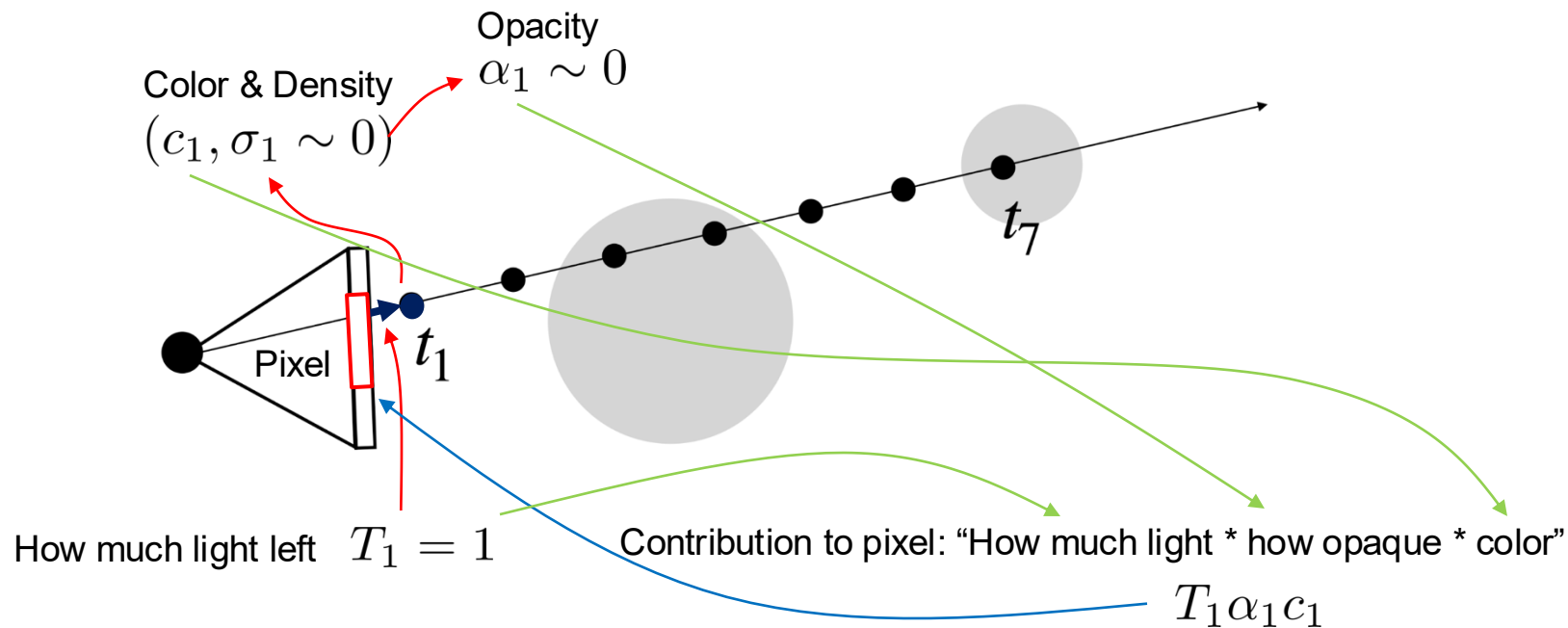
Pixel color



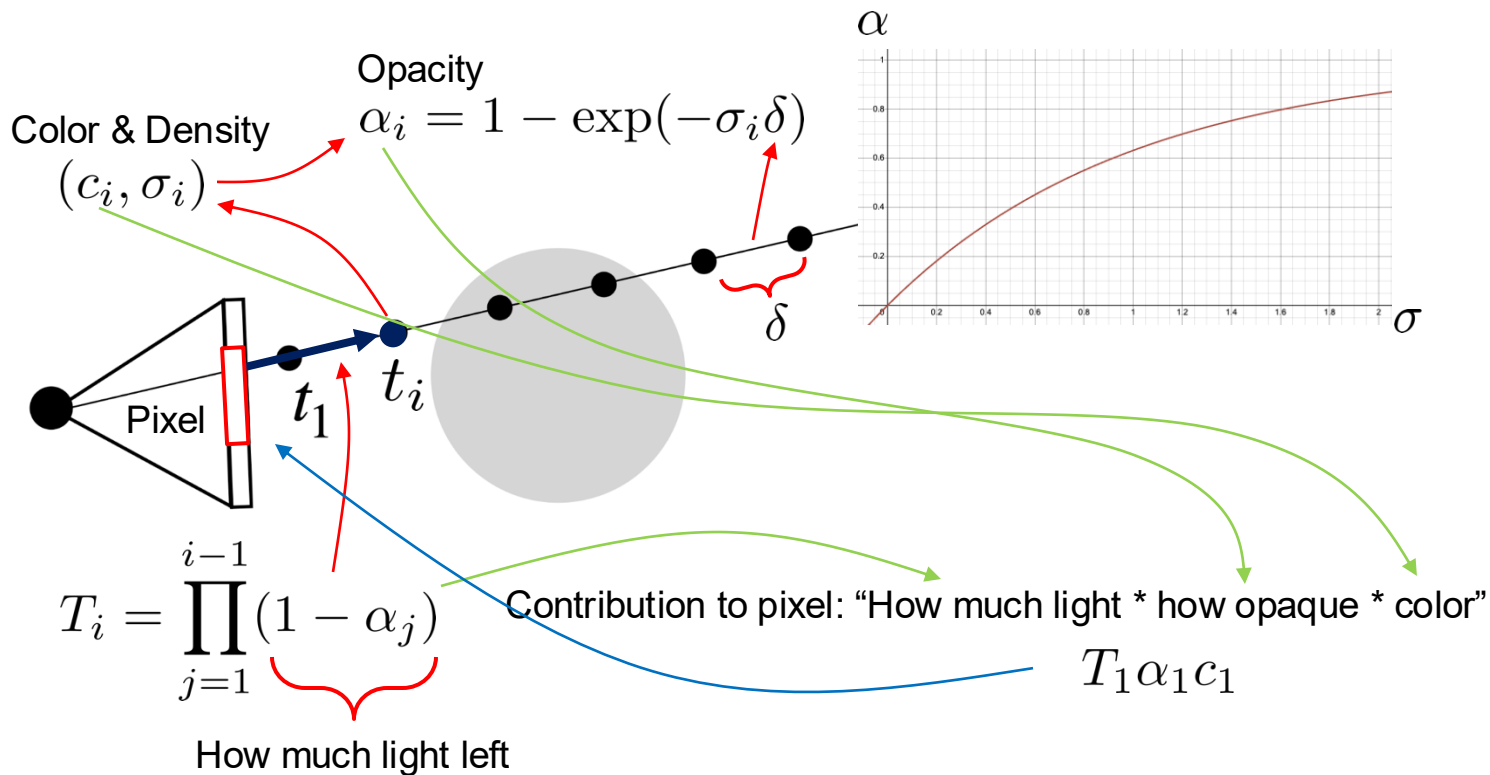
δ, c



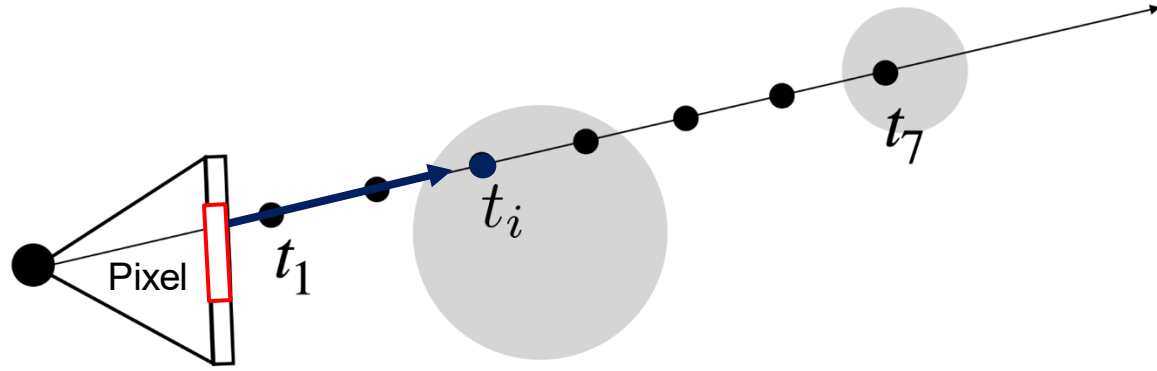
Accumulation of colors



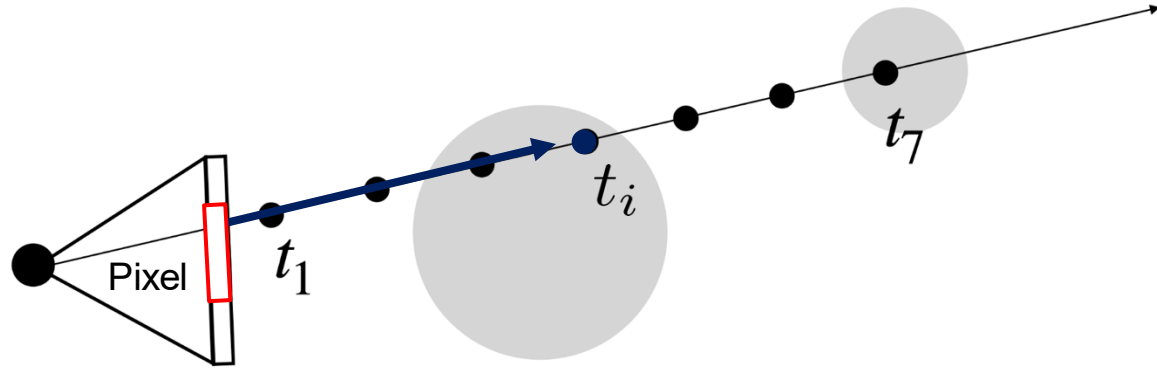
Accumulation of colors



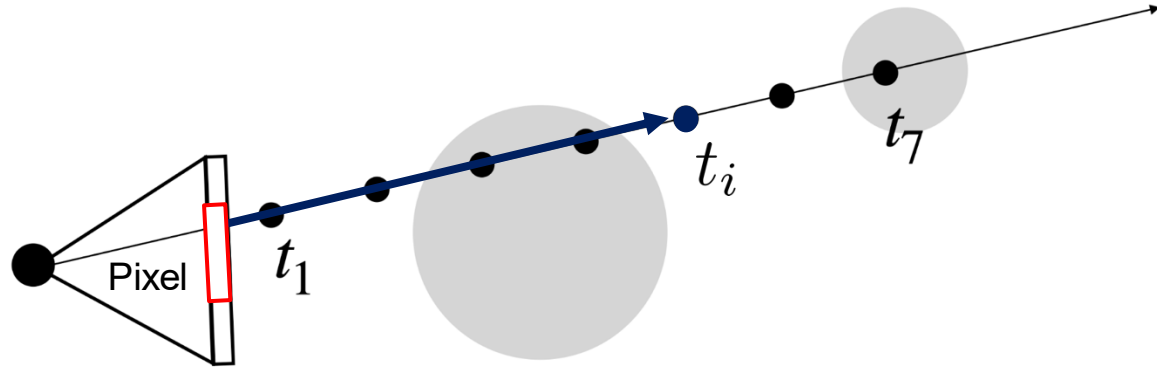
Accumulation of colors



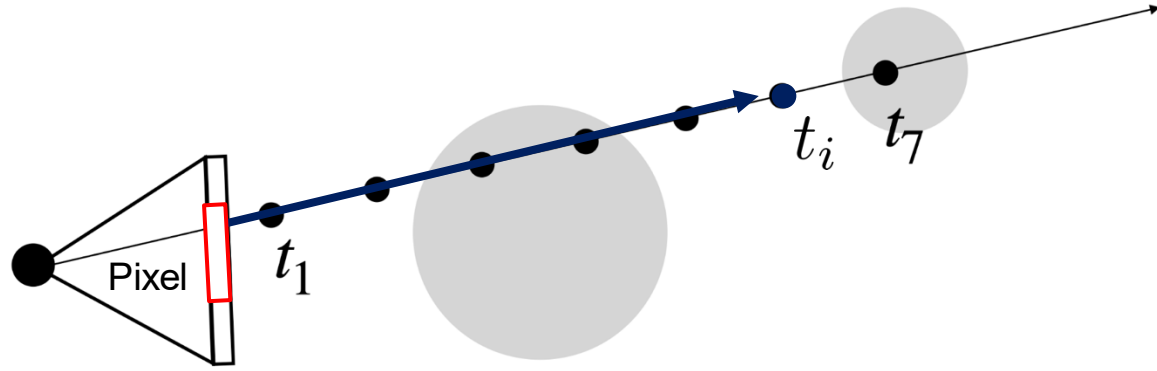
Accumulation of colors



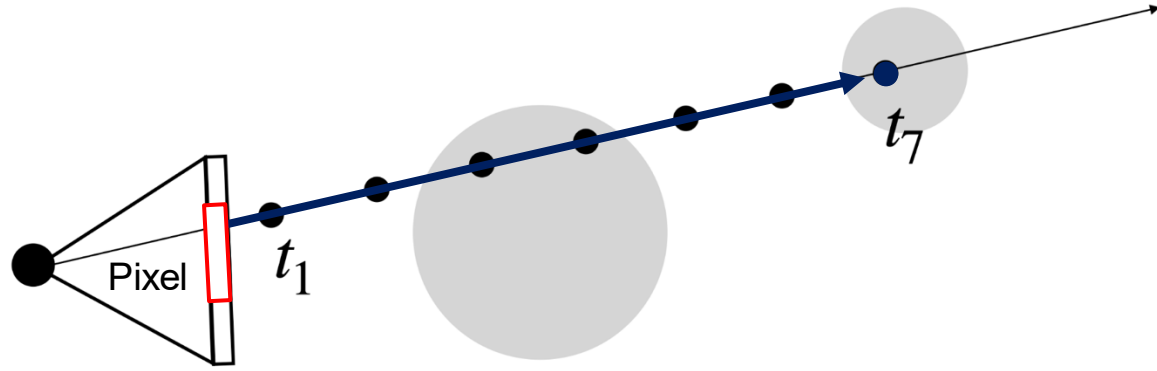
Accumulation of colors



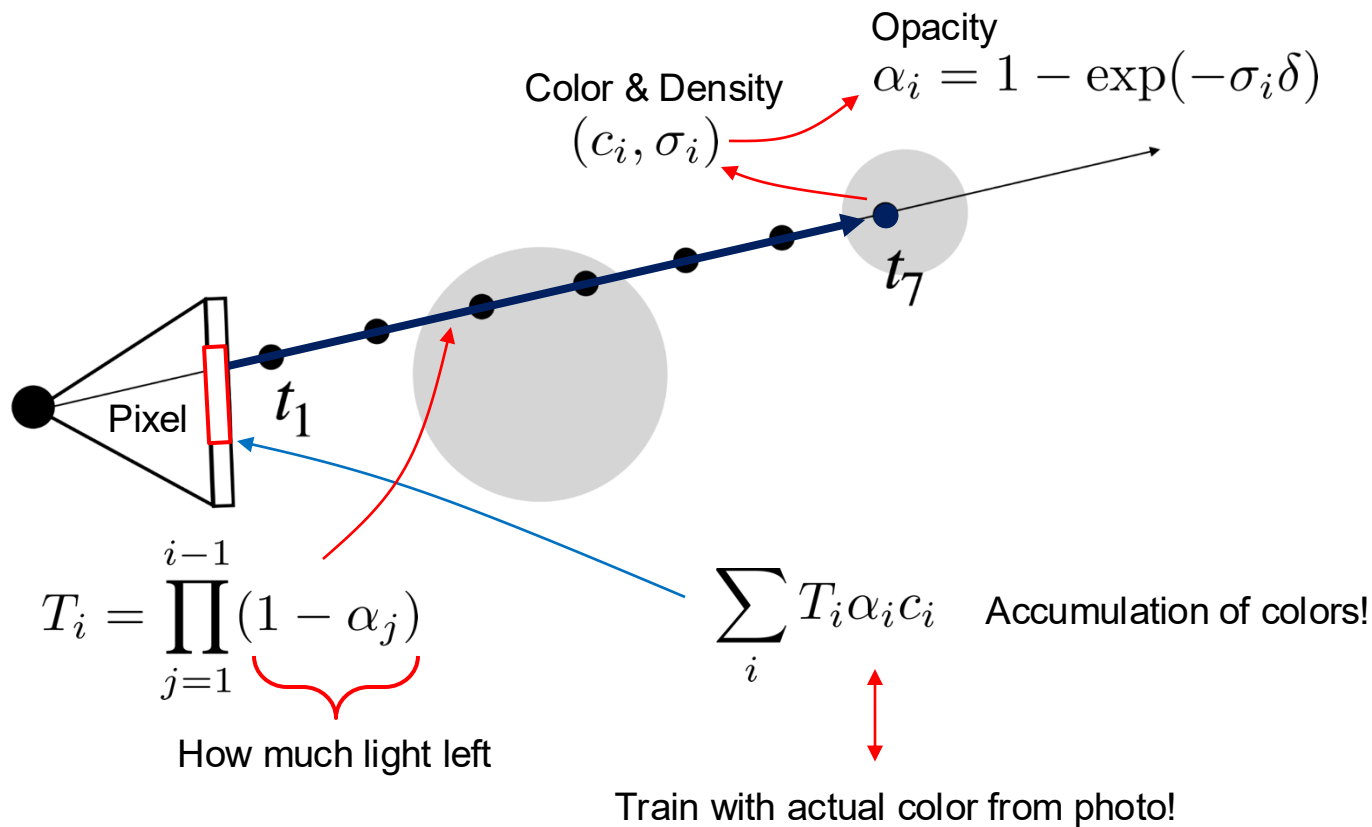
Accumulation of colors



Accumulation of colors



Accumulation of colors



Worksheet :-)